# Problem F
# Question 3: Resource Allocation under Compute Constraints

**Latest Q1-injected Q2 law, cost-family sensitivity, conditional bootstrap, and context analysis**

Final synchronized technical report

24 September 2026

**Numerical source of truth.** Every number, table and plotted quantity in this report is synchronized to the latest successful `python run_all.py` execution supplied by the team. Older report values are superseded.


**Abstract**

The newest Q3 run changes materially because Q2 now injects Q1 quality explicitly. The main quality law uses $G = 1.726229$ and $\gamma = 3.351048$, while the fixed observed mixture has $\Delta_p^* = -0.530858$. At context 4096 and exponential quality cost, the point optima are $(N, D, Q) = (0.1475, 7.5519, 0.6582)$ at $10^{19}$ FLOPs, $(1.1644, 1052.5318, 0.8619)$ at $10^{22}$, and $(27.4340, 5262.2086, 0.9591)$ at $10^{24}$. Quality remains interior at all three main-scenario budgets; the old "saturates by $10^{22}$" conclusion is obsolete. Bootstrap intervals are conditional parameter-perturbation intervals, not a full joint upstream resample.


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## 1 Optimization problem

The 17-domain mixture is fixed to the observed Q1 recipe $p^*$, so this is conditional allocation over $N, D, Q$ rather than unrestricted joint optimization over $p$.

$$C_{train} = 6ND, \qquad C_Q = D[g(Q) - g(Q_0)]_+, \qquad C_{attn} = \eta_c N D L_{ctx},$$

with $Q_0 = 0.5$, $\eta_c = 2 \times 10^{-4}$. The main loss is

$$L = L_0 + 1.726229(1 - Q)^{3.351048} + \omega(N)(-0.530858).$$

## 2 Quality-cost families

Table 1: Latest quality-cost scale audit.

| Family | $g'(0.5)$ | $g'(1)$ | Increment $0.5 \to 1$ |
|---|---|---|---|
| Exponential | $1.2051 \times 10^9$ | $2.4206 \times 10^{10}$ | $3.8334 \times 10^9$ |
| Power | $2.5000 \times 10^9$ | $2.0000 \times 10^{10}$ | $4.6875 \times 10^9$ |
| Logarithmic | $3.3333 \times 10^9$ | $1.8182 \times 10^9$ | $1.2123 \times 10^9$ |

Their absolute scales differ, so differences cannot be attributed to curvature alone.

## 3 Point optimization under three quality-evidence scenarios

Table 2: Latest point solutions at $L_{ctx} = 4096$ using exponential quality cost.

| Scenario | Budget | $N^*$ (B) | $D^*$ (B) | $Q^*$ | Loss |
|---|---|---|---|---|---|
| Main | $10^{19}$ | 0.1475 | 7.5519 | 0.6582 | 2.7205 |
| Main | $10^{22}$ | 1.1644 | 1052.5318 | 0.8619 | 1.9871 |
| Main | $10^{24}$ | 27.4340 | 5262.2086 | 0.9591 | 1.8026 |
| Reverse B8 | $10^{19}$ | 0.1201 | 12.2127 | 0.5000 | 1.6101 |
| Reverse B8 | $10^{22}$ | 0.9195 | 1594.8124 | 0.5000 | 0.9679 |
| Reverse B8 | $10^{24}$ | 26.8095 | 5469.8810 | 0.5000 | 0.7945 |
| No-quality | $10^{19}$ | 0.1201 | 12.2127 | 0.5000 | 2.6177 |
| No-quality | $10^{22}$ | 0.9195 | 1594.8119 | 0.5000 | 1.9755 |
| No-quality | $10^{24}$ | 26.8095 | 5469.8816 | 0.5000 | 1.8020 |

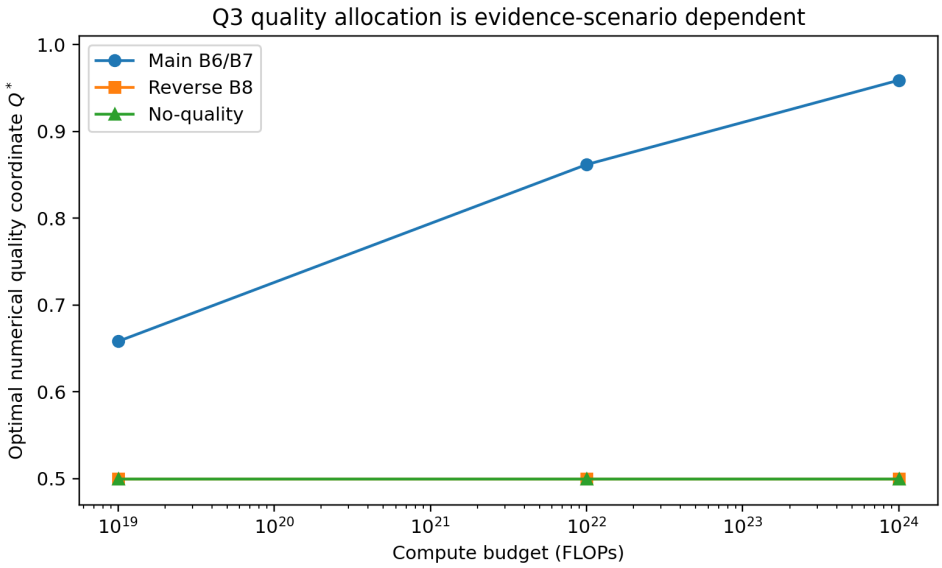



Figure 1: Latest optimal quality coordinate by evidence scenario.

The reverse-B8 absolute loss level is not directly comparable with the main scenario because its quality term has a different sign and reference level; its purpose is structural stress testing.

# 4 Cost-family sensitivity

Table 3: Main-scenario optima under all supplied quality-cost families.

| Cost | Budget | $N^*$ | $D^*$ | $Q^*$ | Loss |
|---|---|---|---|---|---|
| Exponential | $10^{19}$ | 0.1475 | 7.5519 | 0.6582 | 2.7205 |
| Exponential | $10^{22}$ | 1.1644 | 1052.5318 | 0.8619 | 1.9871 |
| Exponential | $10^{24}$ | 27.4340 | 5262.2086 | 0.9591 | 1.8026 |
| Power | $10^{19}$ | 0.1509 | 7.1555 | 0.6075 | 2.7558 |
| Power | $10^{22}$ | 1.2843 | 896.3177 | 0.8581 | 1.9914 |
| Power | $10^{24}$ | 27.6367 | 5197.4551 | 0.9604 | 1.8027 |
| Logarithmic | $10^{19}$ | 0.1657 | 5.7264 | 0.7165 | 2.7385 |
| Logarithmic | $10^{22}$ | 1.0930 | 1171.5585 | 0.9306 | 1.9825 |
| Logarithmic | $10^{24}$ | 27.0628 | 5384.1240 | 0.9847 | 1.8022 |

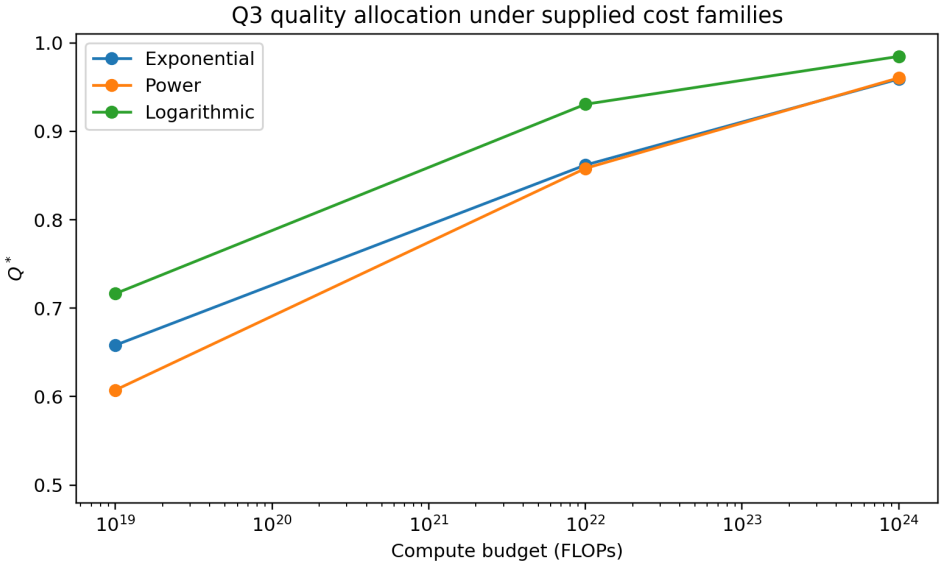



Figure 2: Quality allocation sensitivity to the supplied cost families.

# 5 Conditional uncertainty propagation

The executable labels its bootstrap methodology explicitly as **parameter perturbation**: (1) B1 draw from the Q2 trajectory-cluster bootstrap, (2) jitter of $G, \gamma, H, \delta$, (3) attenuation jitter, and (4) $\Delta_p^*$ jitter. It is **not** a full joint upstream resample.

Table 4: Main-scenario re-optimized 95% intervals from 250 draws.

| Budget | $N$ med | $N$ low | $N$ high | $D$ med | $D$ low | $D$ high | $Q$ med |
|---|---|---|---|---|---|---|---|
| $10^{19}$ | 0.1467 | 0.1270 | 0.1781 | 7.60 | 6.37 | 8.83 | 0.6580 |
| $10^{22}$ | 1.1797 | 0.5209 | 2.7090 | 1038.47 | 492.88 | 2061.38 | 0.8604 |
| $10^{24}$ | 27.3986 | 19.5462 | 32.9721 | 5271.24 | 4390.21 | 7341.00 | 0.9582 |

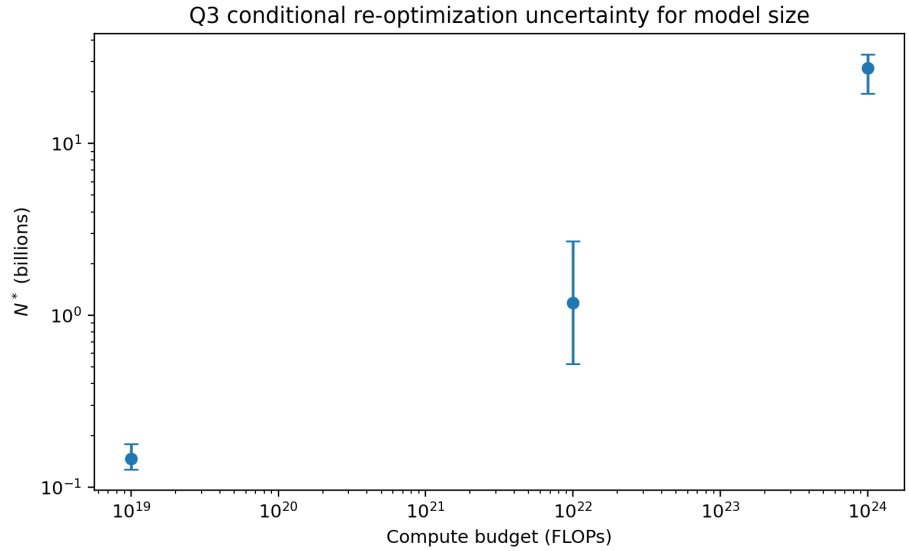



Figure 3: Conditional re-optimization interval for $N^*$.

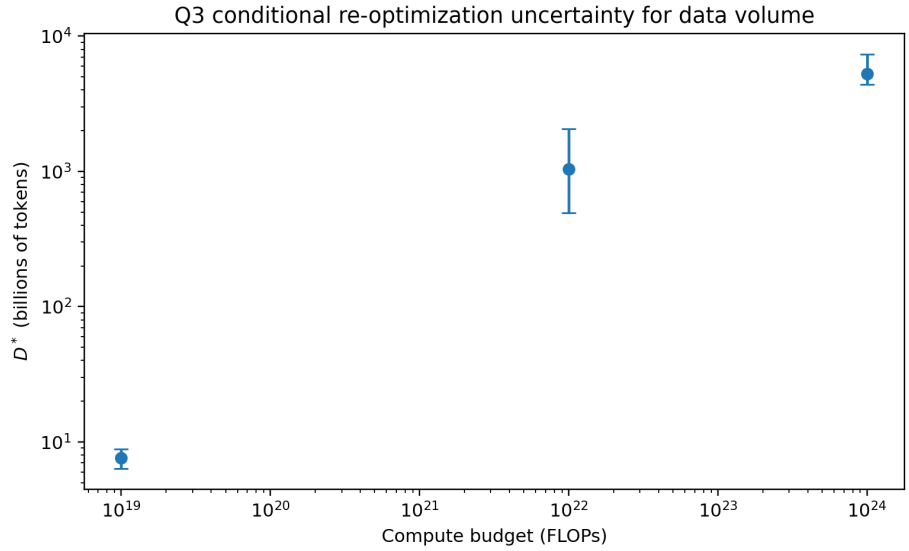



Figure 4: Conditional re-optimization interval for $D^*$.

All 250 main-scenario draws are interior in $Q$ at $10^{19}, 10^{22}, 10^{24}$. Reverse-B8 and no-quality draws remain at $Q_0$ for all three budgets.

# 6 Transition budget

The newest transition sweep reports only the leave-$Q_0$ threshold:

$$\boxed{C_{leave} = 3.162 \times 10^{18} \text{ FLOPs}}, \qquad 95\% \text{ reported interval} = [3.162, 3.162] \times 10^{18}.$$

The report reproduces this degenerate interval exactly but does not interpret it as proof of zero uncertainty. No new "reach $Q = 1$" threshold is printed by the latest executable, and the main solution is still interior ($Q^* = 0.9591$) at $10^{24}$, so the obsolete saturation-threshold table is removed.

# 7 Context sensitivity

C7 provides the five evaluated contexts 2048, 4096, 8192, 32768 and 131072. The accounting crossover is

$$L_{ctx}^{cross} = 6/\eta_c = 30,000 \text{ tokens},$$

which is a cost-accounting boundary, not a performance threshold.

Table 5: Latest context sensitivity at $10^{22}$ FLOPs.

| Context | $N^*$ | $D^*$ | $Q^*$ | Loss | Attn.% | Train% |
|---|---|---|---|---|---|---|
| 2048 | 1.2130 | 1071.8555 | 0.8613 | 1.9845 | 5.33 | 78.01 |
| 4096 | 1.1644 | 1052.5318 | 0.8619 | 1.9871 | 10.04 | 73.53 |
| 8192 | 1.0843 | 1014.6846 | 0.8630 | 1.9918 | 18.03 | 66.01 |
| 32768 | 0.8306 | 828.7478 | 0.8692 | 2.0136 | 45.11 | 41.30 |
| 131072 | 0.5669 | 498.2579 | 0.8838 | 2.0610 | 74.04 | 16.95 |

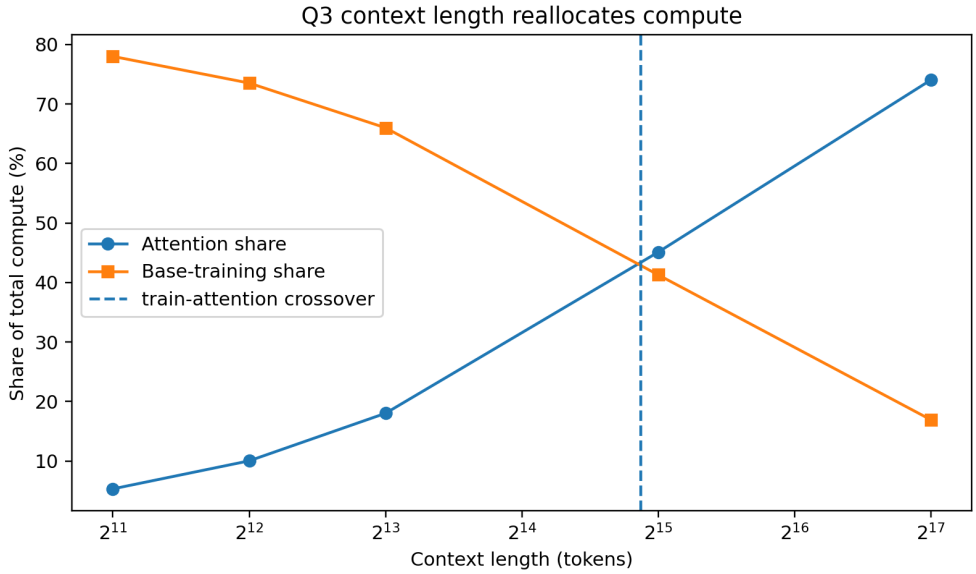



Figure 5: Longer context reallocates compute toward the supplied attention-cost proxy.

## 8 Known limitations - honest disclosure

**Q3 limitations.** (i) Optimization is conditional on the fixed observed recipe $p^*$; it is not unrestricted joint optimization over the 17-simplex. (ii) Cost-family comparisons are conditional on the supplied absolute normalizations as well as curvature. (iii) Bootstrap uncertainty is parameter/fit perturbation with re-optimization, not a full joint upstream resample. (iv) The high-budget optima omit hardware parallelism, communication, memory, wall-clock, optimizer-stability and data-availability caps. (v) The latest executable does not print a fresh dense-grid/KKT global-optimality audit, so this report does not claim one.

## 9 Final Q3 answer

With the latest Q1-injected Q2 law, quality investment remains interior over all three official budgets instead of saturating by $10^{22}$. The robust message is therefore budget- and evidence-dependent allocation: more compute increases $N$, $D$ and, under the main scenario, $Q$, while longer context crowds out the other resources through the specified attention-cost accounting.